Assignment

1. You open the Heart Disease dataset in Jupyter Notebook. What is the difference between a .py file and a .ipynb file?

Answer:

- **.py file:** A standard Python script file that contains only Python code and is usually run as a complete program.
- **.ipynb file:** A Jupyter Notebook file that contains code, output, text, images, and visualizations in separate cells, allowing interactive analysis and documentation.

Example: A .py file is used for writing Python programs, while a .ipynb file is used for data analysis, machine learning, and interactive coding in Jupyter Notebook.

2. The dataset has columns like age, cholesterol, and sex. What is the difference between a numerical feature and a categorical feature?

Answer:

- **Numerical Feature:** Contains numeric values that can be measured or counted. Examples: age, cholesterol, blood pressure.
- **Categorical Feature:** Contains labels or categories that represent groups. Examples: sex (Male/Female), chest pain type, disease status.

Example: In the Heart Disease dataset, age and cholesterol are numerical features, while sex is a categorical feature.

3. Why does the data type of the target variable matter?

Answer:
The data type of the target variable matters because it helps determine the type of prediction problem and the machine learning algorithm to use. If the target variable is an integer (int) representing categories (e.g., 0 = No Disease, 1 = Disease), it is treated as a classification problem. If it represents continuous numeric values, it is treated as a regression problem.

Example: In the Heart Disease dataset, a target column with values 0 and 1 indicates a classification task.

4. You compute the correlation between age and heart disease presence and get 0.23. What does a correlation value tell you about two variables?

Answer:
A correlation value shows the strength and direction of the relationship between two variables.

- A value close to +1 indicates a strong positive relationship.
- A value close to -1 indicates a strong negative relationship.
- A value close to 0 indicates little or no linear relationship.

Since the correlation between age and heart disease presence is 0.23, it indicates a weak positive relationship, meaning that as age increases, the likelihood of heart disease tends to increase slightly, but the relationship is not strong.

5. A hospital rule says: if cholesterol > 240, flag as high risk. What is the fundamental difference between a rule-based system and a data-driven approach?

Answer:
The fundamental difference is that a rule-based system uses predefined rules created by humans (e.g., if cholesterol > 240, then high risk), while a data-driven approach learns patterns automatically from data and makes predictions based on those learned patterns.

Example: A rule-based system follows fixed conditions, whereas a machine learning model can analyze multiple factors such as age, cholesterol, blood pressure, and sex to predict heart disease risk.